

Algebra I Written Exam

2026 James Clemens Math Tournament

INSTRUCTIONS:

1. You will have 90 minutes to complete this examination.
2. This exam consists of 25 multiple choice questions and 3 tiebreaker questions. The multiple choice questions are each worth 4 points if answered correctly and -1 points if answered incorrectly. There is no penalty for unanswered questions. The tiebreaker questions are worth 0.1 points if answered correctly and 0 points if answered incorrectly or left unanswered.
3. Use a #2 pencil.
4. Calculators, cellular devices, books, notes, and other aids are prohibited during this exam. Scratch paper will be provided for calculations. If you require additional scratch paper, simply raise your hand and your proctor will assist you.
5. Mark your answers to the questions in the Scantron form provided. You may use the test booklets for scratch work, but only the answers marked in the Scantron form will be counted. Make sure to fill in the bubbles on the Scantron form **completely** and erase any stray marks.
6. Write your answers to the tiebreaker questions on the back of the Scantron form. Label the tiebreakers as TB1, TB2, and TB3.
7. NOTA denotes “None Of The Above.”
8. Diagrams are not necessarily drawn to scale.
9. In the event of a tie, answers will be evaluated backwards starting from multiple choice question 25 to determine a winner.
10. Although this math tournament is intended to demonstrate your knowledge and skills in math, be sure to enjoy the experience and have fun!
11. You may keep the test booklet after the exam.

DO NOT TURN THIS PAGE UNTIL TOLD TO DO SO

- What is the sum of the first 20 natural numbers?
(A) 21 (B) 105 (C) 210 (D) 420 (E) NOTA
- John has 63 pizza slices and x friends. What is the smallest value of x such that John can distribute an equal amount of pizza slices to each friend?
(A) 2 (B) 3 (C) 7 (D) 9 (E) NOTA
- Jeffrey is practicing his serves in tennis! He records the speed, in miles per hour, of five serves to be 86, 89, 90, 97, and 78. What is the average speed of his serves, in mile per hour?
(A) 86 (B) 87 (C) 88 (D) 89 (E) NOTA
- Evaluate the sum of all numbers in the arithmetic sequence $-99, -91, -83, \dots, 93, 101$.
(A) 26 (B) 2600 (C) 1352 (D) -52 (E) NOTA
- Determine the value of $\sqrt{9 + 9\sqrt{9 + 9\sqrt{9}}}$.
(A) 3 (B) $3\sqrt{7}$ (C) 9 (D) $9\sqrt{3}$ (E) NOTA
- Find the 2026th letter in JAMESCLEMENSJAMESCLEMENS....
(A) S (B) J (C) M (D) E (E) NOTA
- What is the coefficient of the x^3 term in the expansion of $(x - 3)(x^2 + 5)(2x + 3)$?
(A) -15 (B) -3 (C) 1 (D) 2 (E) NOTA
- A loaded dice has the property that the probability of rolling an even number is equivalent to the reciprocal of the even number (for example, there is a $\frac{1}{2}$ chance of rolling a 2). What is the probability of rolling an odd number?
(A) $\frac{1}{3}$ (B) $\frac{5}{12}$ (C) $\frac{1}{8}$ (D) $\frac{1}{12}$ (E) NOTA
- Let the number of unique arrangements of the letters in JAMESCLEMENS be expressed as $\frac{a!}{b!c!d!}$, where b , c , and d sum to 7. Find $\frac{a}{bcd}$.
(A) 12 (B) 144 (C) 1 (D) 0.5 (E) NOTA

10. A palindrome is a number whose value remains the same when written backward. How many five-digit palindromes are there?
- (A) 900 (B) 100000 (C) 50 (D) 0 (E) NOTA
11. Given that the intersection between the line $2x + 3y = 8$ and the line perpendicular to it that runs through the point $(3, 5)$ is (x, y) , find $x + y$.
- (A) 3 (B) 12 (C) 5 (D) 14 (E) NOTA
12. Find the area of the region enclosed by the graphs of $x = 0$, $y = 0$, $y = -x + 3$, and $y = -x + 5$.
- (A) 5 (B) 8 (C) 16 (D) 24 (E) NOTA
13. Find the smallest natural number that is not a divisor of 5040.
- (A) 1 (B) 7 (C) 8 (D) 9 (E) NOTA
14. How many ways can 5 people sit at a circular table with 7 seats? Two arrangements are considered the same if one arrangement can be rotated to obtain the other.
- (A) 42 (B) 720 (C) 360 (D) 504 (E) NOTA
15. Let $a \triangle b = 3(a + b)$ and $a \nabla b = 4ab$. If $x \triangle y = 75$ and $x \nabla y = 624$, determine the value of $(x - y)^2$.
- (A) 192 (B) 16 (C) 9 (D) 1 (E) NOTA
16. A bookshelf contains 12 books: 5 math books, 2 English books, 3 history books, and 2 science books. Given that all books of the same topic are indistinguishable, in how many unique ways can you select any number of books such that you have at least three different topics?
- (A) 720 (B) 152 (C) 60 (D) 180 (E) NOTA
17. Jeffrey is back with his serves! This time, he wants to train his aim. He sets up a circular target inscribed in a 10 foot by 10 foot square, followed by another square inscribed within the circle, then another circular target inscribed within the smaller square, and so on until he's set up 5 circular targets in this pattern. Given that Jeffrey is guaranteed to hit within the 10 by 10 feet box, what is the probability he hits within the third largest square target?
- (A) $\frac{1}{8}$ (B) $\frac{1}{4}$ (C) $\frac{1}{2}$ (D) $\frac{3}{4}$ (E) NOTA

23. Let

$$f(x) = \begin{cases} 3x + 1, & \text{if } x \text{ is odd} \\ \frac{x}{2}, & \text{if } x \text{ is even} \end{cases}$$

and let $f^{(n)}(x)$ denote f applied to x , n times — that is,

$$f^{(n)}(x) = \underbrace{f(f(\dots f(x) \dots))}_{n \text{ times}}.$$

Determine $f^{(20)}(26)$.

- (A) 26 (B) 4 (C) 2 (D) 1 (E) NOTA

24. The repeating portion of the decimal expansion of a fraction $\frac{a}{b}$ has the unique property that when multiplied by 2, 3, 4, 5, or 6, the number remains a permutation of its original digits, and when multiplied by b , the number equals 999999. Find the smallest value of $a + b$ such that a and b are both positive integers.

- (A) 8 (B) 27 (C) 51 (D) 16 (E) NOTA

25. Let $f(x) = x^2 + 10x - 17$ and let $g(x) = 2x^2 - 5x - 53$. The graphs of $y = f(x)$ and $y = g(x)$ intersect at two points A and B . What is the sum of the y -coordinates of A and B ?

- (A) 15 (B) 36 (C) 269 (D) 413 (E) NOTA

Tiebreakers:

TB1. How many positive integers x satisfy the inequality $12xy + 3y \leq 6xy + 51y$?

TB2. How many numbers are in the sequence

$$3\frac{1}{4}, 4\frac{1}{2}, 5\frac{3}{4}, \dots, 33\frac{1}{4},$$

inclusive?

TB3. How many integers less than 10000 can be expressed as the sum of any number of perfect fourths?